Introduction to Number Theory

Number theory is about **integers** and their properties.

We will start with the basic principles of

- divisibility,
- greatest common divisors,
- least common multiples, and
- modular arithmetic

and look at some relevant algorithms.

Division

If a and b are integers with $a \neq 0$, we say that a **divides** b if there is an integer c so that b = ac.

When a divides b we say that a is a **factor** of b and that b is a **multiple** of a.

The notation **a | b** means that a divides b.

We write **a X b** when a does not divide b (see book for correct symbol).

Divisibility Theorems

For integers a, b, and c it is true that

- if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$
  Example: $3 \mid 6$ and $3 \mid 9$, so $3 \mid 15$.
- if $a \mid b$, then $a \mid bc$ for all integers c
  Example: $5 \mid 10$, so $5 \mid 20$, $5 \mid 30$, $5 \mid 40$, …
- if $a \mid b$ and $b \mid c$, then $a \mid c$
  Example: $4 \mid 8$ and $8 \mid 24$, so $4 \mid 24$.

Primes

A positive integer p greater than 1 is called prime if the only positive factors of p are 1 and p.

A positive integer that is greater than 1 and is not prime is called composite.

The fundamental theorem of arithmetic:

Every positive integer can be written **uniquely** as the **product of primes**, where the prime factors are written in order of increasing size.

Primes

Examples:

$15 =$	$3 \cdot 5$
$48 =$	$2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 = 2^4 \cdot 3$
$17 =$	17
$100 =$	$2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
$512 =$	$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^9$
$515 =$	$5 \cdot 103$
$28 =$	$2 \cdot 2 \cdot 7$

Primes

If n is a composite integer, then n has a prime divisor less than or equal $\sqrt{n}$.

This is easy to see: if n is a composite integer, it must have two prime divisors p_1 and p_2 such that $p_1 \cdot p_2 = n$.

p_1 and p_2 cannot both be greater than $\sqrt{n}$, because then $p_1 \cdot p_2 > n$.

The Division Algorithm

Let **a** be an integer and **d** a positive integer. Then there are unique integers **q** and **r**, with $0 \leq r < d$, such that $a = dq + r$.

In the above equation,

- **d** is called the divisor,
- **a** is called the dividend,
- **q** is called the quotient, and
- **r** is called the remainder.

The Division Algorithm

Example:

When we divide 17 by 5, we have

$17 = 5 \cdot 3 + 2.$

- 17 is the dividend,
- 5 is the divisor,
- 3 is called the quotient, and
- 2 is called the remainder.

The Division Algorithm

Another example:

What happens when we divide -11 by 3 ?

Note that the remainder cannot be negative.

$-11 = 3 \cdot (-4) + 1.$

- -11 is the dividend,
- 3 is the divisor,
- -4 is called the quotient, and
- 1 is called the remainder.

Greatest Common Divisors

Let a and b be integers, not both zero.
The largest integer d such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor** of a and b.

The greatest common divisor of a and b is denoted by gcd(a, b).

Example 1: What is gcd(48, 72) ?

The positive common divisors of 48 and 72 are 1, 2, 3, 4, 6, 8, 12, 16, and 24, so gcd(48, 72) = 24.

Example 2: What is gcd(19, 72) ?

The only positive common divisor of 19 and 72 is 1, so gcd(19, 72) = 1.

Greatest Common Divisors

Using prime factorizations:

$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}$, $b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$,

where $p_1 < p_2 < \dots < p_n$ and $a_i, b_i \in \mathbf{N}$ for $1 \le i \le n$

$gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \dots p_n^{\min(a_n, b_n)}$

Example:

$a = 60 = 2^2\ 3^1\ 5^1$

$b = 54 = 2^1\ 3^3\ 5^0$

$gcd(a, b) = 2^1\ 3^1\ 5^0 = 6$

Relatively Prime Integers

Definition:

Two integers a and b are **relatively prime** if gcd(a, b) = 1.

Examples:

Are 15 and 28 relatively prime?
Yes, gcd(15, 28) = 1.
Are 55 and 28 relatively prime?
Yes, gcd(55, 28) = 1.
Are 35 and 28 relatively prime?
No, gcd(35, 28) = 7.

Relatively Prime Integers

Definition:

The integers $a_1, a_2, \ldots, a_n$ are **pairwise relatively prime** if $\gcd(a_i, a_j) = 1$ whenever $1 \le i < j \le n$.

Examples:

Are 15, 17, and 27 pairwise relatively prime?
No, because gcd(15, 27) = 3.

Are 15, 17, and 28 pairwise relatively prime?
Yes, because gcd(15, 17) = 1, gcd(15, 28) = 1 and gcd(17, 28) = 1.

Least Common Multiples

Definition:

The **least common multiple** of the positive integers a and b is the smallest positive integer that is divisible by both a and b.

We denote the least common multiple of a and b by lcm(a, b).

Examples:

lcm(3, 7) = 21

lcm(4, 6) = 12

lcm(5, 10) = 10

Least Common Multiples

Using prime factorizations:

$a = p_1^{a_1} p_2^{a_2} \ldots p_n^{a_n}$, $b = p_1^{b_1} p_2^{b_2} \ldots p_n^{b_n}$,

where $p_1 < p_2 < \ldots < p_n$ and $a_i, b_i \in \mathbf{N}$ for $1 \leq i \leq n$

$\mathrm{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \ldots p_n^{\max(a_n, b_n)}$

Example:

$a = 60 = 2^2\ 3^1\ 5^1$

$b = 54 = 2^1\ 3^3\ 5^0$

$\mathrm{lcm}(a, b) = 2^2\ 3^3\ 5^1 = 4 \cdot 27 \cdot 5 = 540$

GCD and LCM

$a = 60 = 2^2 \; 3^1 \; 5^1$

$b = 54 = 2^1 \; 3^3 \; 5^0$

$\gcd(a, b) = 2^1 \; 3^1 \; 5^0 = 6$

$\text{lcm}(a, b) = 2^2 \; 3^3 \; 5^1 = 540$

Theorem: $a \cdot b = \gcd(a,b) \cdot \text{lcm}(a,b)$

Modular Arithmetic

Let a be an integer and m be a positive integer. We denote by **a mod m** the remainder when a is divided by m.

Examples:

9 mod 4 = 1

9 mod 3 = 0

9 mod 10 = 9

-13 mod 4 = 3

Congruences

Let a and b be integers and m be a positive integer. We say that **a is congruent to b modulo m** if m divides a – b.

We use the notation **$a \equiv b \pmod{m}$** to indicate that a is congruent to b modulo m.

In other words:
$a \equiv b \pmod{m}$ if and only if **a mod m = b mod m**.

Congruences

Examples:

Is it true that $46 \equiv 68 \pmod{11}$?

Yes, because $11 \mid (46 - 68)$.

Is it true that $46 \equiv 68 \pmod{22}$?

Yes, because $22 \mid (46 - 68)$.

For which integers z is it true that $z \equiv 12 \pmod{10}$?

It is true for any $z \in \{\ldots, -28, -18, -8, 2, 12, 22, 32, \ldots\}$

Theorem: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Congruences

Theorem: Let m be a positive integer.
If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then
$a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof:
We know that $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ implies that there are integers s and t with
$b = a + sm$ and $d = c + tm$.

Therefore,
$b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
$bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$.

Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

The Euclidean Algorithm

The **Euclidean Algorithm** finds the **greatest common divisor** of two integers a and b.

For example, if we want to find gcd(287, 91), we **divide** 287 by 91:

$287 = 91 \cdot 3 + 14$

We know that for integers a, b and c,
if a | b and a | c, then a | (b + c).

Therefore, any divisor of 287 and 91 must also be a divisor of $287 - 91 \cdot 3 = 14$.

Consequently, gcd(287, 91) = gcd(14, 91).

The Euclidean Algorithm

In the next step, we divide 91 by 14:

$91 = 14 \cdot 6 + 7$

This means that gcd(14, 91) = gcd(14, 7).

So we divide 14 by 7:

$14 = 7 \cdot 2 + 0$

We find that 7 | 14, and thus gcd(14, 7) = 7.

Therefore, gcd(287, 91) = 7.

The Euclidean Algorithm

In **pseudocode**, the algorithm can be implemented as follows:

```
procedure gcd(a, b: positive integers)
x := a
y := b
while y ≠ 0
begin
	r := x mod y
	x := y
	y := r
end {x is gcd(a, b)}
```

Representations of Integers

Let b be a positive integer greater than 1.
Then if n is a positive integer, it can be expressed **uniquely** in the form:

$n = a_k b^k + a_{k-1} b^{k-1} + \ldots + a_1 b + a_0,$

where k is a nonnegative integer,
$a_0, a_1, \ldots, a_k$ are nonnegative integers less than b,
and $a_k \neq 0$.

Example for b=10:

$859 = 8 \cdot 10^2 + 5 \cdot 10^1 + 9 \cdot 10^0$

Representations of Integers

Example for b=2 (binary expansion):

$(10110)_2 = 1 \cdot 2^4 + 1 \cdot 2^2 + 1 \cdot 2^1 = (22)_{10}$

Example for b=16 (hexadecimal expansion):

(we use letters A to F to indicate numbers 10 to 15)

$(3A0F)_{16} = 3 \cdot 16^3 + 10 \cdot 16^2 + 15 \cdot 16^0 = (14863)_{10}$

Representations of Integers

How can we construct the base b expansion of an integer n?

First, divide n by b to obtain a quotient q_0 and remainder a_0, that is,

$n = bq_0 + a_0$, where $0 \leq a_0 < b$.

The remainder a_0 is the rightmost digit in the base b expansion of n.

Next, divide q_0 by b to obtain:

$q_0 = bq_1 + a_1$, where $0 \leq a_1 < b$.

a_1 is the second digit from the right in the base b expansion of n. Continue this process until you obtain a quotient equal to zero.

Representations of Integers

Example:
What is the base 8 expansion of $(12345)_{10}$?

First, divide 12345 by 8:
$12345 = 8 \cdot 1543 + 1$

$1543 = 8 \cdot 192 + 7$
$192 = 8 \cdot 24 + 0$
$24 = 8 \cdot 3 + 0$
$3 = 8 \cdot 0 + 3$

The result is: $(12345)_{10} = (30071)_8$.

Representations of Integers

```
procedure base_b_expansion(n, b: positive integers)
q := n
k := 0
while q ≠ 0
begin
      a_k := q mod b
      q := ⌊q/b⌋
      k := k + 1
end
{the base b expansion of n is (a_{k-1} … a_1a_0)_b}
```

Addition of Integers

Let $a = (a_{n-1}a_{n-2}\ldots a_1a_0)_2$, $b = (b_{n-1}b_{n-2}\ldots b_1b_0)_2.$

How can we add these two binary numbers?

First, add their rightmost bits:

$a_0 + b_0 = c_0 \cdot 2 + s_0,$

where s_0 is the **rightmost bit** in the binary expansion of $a + b$, and c_0 is the **carry**.

Then, add the next pair of bits and the carry:

$a_1 + b_1 + c_0 = c_1 \cdot 2 + s_1,$

where s_1 is the **next bit** in the binary expansion of $a + b$, and c_1 is the carry.

Addition of Integers

Continue this process until you obtain c_{n-1}.

The leading bit of the sum is $s_n = c_{n-1}$.

The result is:

$a + b = (s_n s_{n-1} \ldots s_1 s_0)_2$

Addition of Integers

Example:

Add $a = (1110)_2$ and $b = (1011)_2$.

$a_0 + b_0 = 0 + 1 = 0 \cdot 2 + 1$, so that $c_0 = 0$ and $s_0 = 1$.

$a_1 + b_1 + c_0 = 1 + 1 + 0 = 1 \cdot 2 + 0$, so $c_1 = 1$ and $s_1 = 0$.

$a_2 + b_2 + c_1 = 1 + 0 + 1 = 1 \cdot 2 + 0$, so $c_2 = 1$ and $s_2 = 0$.

$a_3 + b_3 + c_2 = 1 + 1 + 1 = 1 \cdot 2 + 1$, so $c_3 = 1$ and $s_3 = 1$.

$s_4 = c_3 = 1$.

Therefore, $s = a + b = (11001)_2$.

Addition of Integers

How do we (humans) add two integers?

$$
\begin{array}{rl}
\text{Example:} \quad & \begin{array}{r} \color{red}{1\,1\,1} \quad \color{red}{\text{carry}} \\ 7583 \\ +\ 4932 \\ \hline 12515 \end{array}
\end{array}
$$

$$
\begin{array}{rl}
\text{Binary expansions:} \quad & \begin{array}{r} \color{red}{1\ \ 1} \quad \color{red}{\text{carry}} \\ (1011)_2 \\ +\ (1010)_2 \\ \hline (10101)_2 \end{array}
\end{array}
$$

Addition of Integers

Let $a = (a_{n-1}a_{n-2}\ldots a_1a_0)_2$, $b = (b_{n-1}b_{n-2}\ldots b_1b_0)_2$.

How can we **algorithmically** add these two binary numbers?

First, add their rightmost bits:

$a_0 + b_0 = c_0 \cdot 2 + s_0$,

where s_0 is the **rightmost bit** in the binary expansion of a + b, and c_0 is the **carry**.

Then, add the next pair of bits and the carry:

$a_1 + b_1 + c_0 = c_1 \cdot 2 + s_1$,

where s_1 is the **next bit** in the binary expansion of a + b, and c_1 is the carry.

Addition of Integers

Continue this process until you obtain c_{n-1}.

The leading bit of the sum is $s_n = c_{n-1}$.

The result is:

$a + b = (s_n s_{n-1} \ldots s_1 s_0)_2$

Addition of Integers

Example:

Add $a = (1110)_2$ and $b = (1011)_2$.

$a_0 + b_0 = 0 + 1 = 0 \cdot 2 + 1$, so that $c_0 = 0$ and $s_0 = 1$.

$a_1 + b_1 + c_0 = 1 + 1 + 0 = 1 \cdot 2 + 0$, so $c_1 = 1$ and $s_1 = 0$.

$a_2 + b_2 + c_1 = 1 + 0 + 1 = 1 \cdot 2 + 0$, so $c_2 = 1$ and $s_2 = 0$.

$a_3 + b_3 + c_2 = 1 + 1 + 1 = 1 \cdot 2 + 1$, so $c_3 = 1$ and $s_3 = 1$.

$s_4 = c_3 = 1$.

Therefore, $s = a + b = (11001)_2$.

Addition of Integers

procedure add(a, b: positive integers)
$c := 0$
for $j := 0$ to $n-1$
begin
    $d := \lfloor (a_j + b_j + c)/2 \rfloor$
    $s_j := a_j + b_j + c - 2d$
    $c := d$
end
$s_n := c$
{the binary expansion of the sum is $(s_n s_{n-1} \ldots s_1 s_0)_2$}